

FORTIGATE NSE4

Lab 7 — Automation Stitches

1. What an Automation Stitch Is

An automation stitch binds an event to a response. A trigger defines the condition the FortiGate watches for; one or more actions define what happens when that condition occurs. Once the stitch is enabled, the response runs without an administrator present.

Stitches are configured under **Security Fabric > Automation**.



Figure 1 — Automation sits under the Security Fabric menu

2. Creating a Stitch



Figure 2 — Stitch header: name, status, action execution mode, and description

Field	Purpose
Name	Identifies the stitch. Worth naming for the outcome rather than the trigger, so the intent is readable in the list
Status	Enable or Disable. A stitch can be built and left disabled until it has been reviewed, which is the safer sequence for anything with a disruptive action
Action execution	Sequential or Parallel. Determines how multiple actions are ordered
Description	Free text. The place to record why the stitch exists — the field that saves the next administrator from guessing

► 2.1 Sequential Versus Parallel

Mode	Behaviour	Consequence
Sequential	Actions run one after another, in the order listed, with an optional delay between each	A failed action stops the chain — the actions after it never run. Ordering and timing can be controlled
Parallel	All actions are dispatched simultaneously	No delays and no ordering. A failure in one action does not affect the others

The choice is not cosmetic. Sequential is correct when actions depend on each other or on timing — quarantine a host, wait, then verify the quarantine took effect. Parallel is correct when actions are independent and speed matters, such as notifying several channels at once.

3. Triggers

The trigger is the condition being watched. FortiOS groups the available triggers by source.

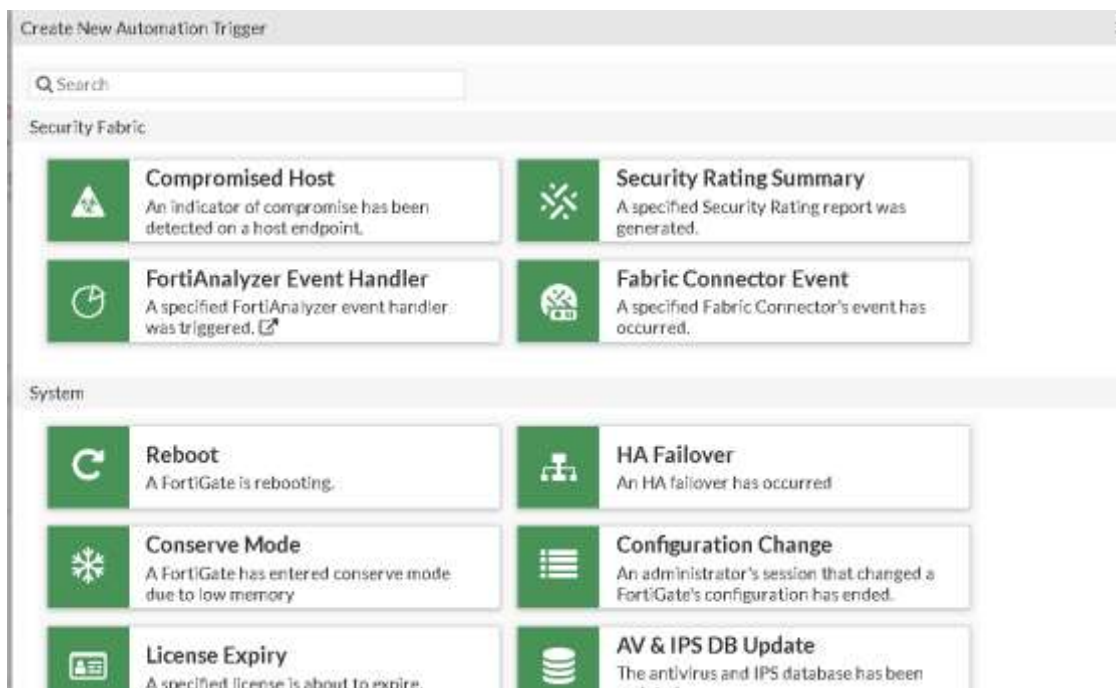


Figure 3 — Available automation triggers, grouped by Security Fabric and System

4. Actions

Actions define the response. They fall into two broad categories: those that contain a threat, and those that inform a person.

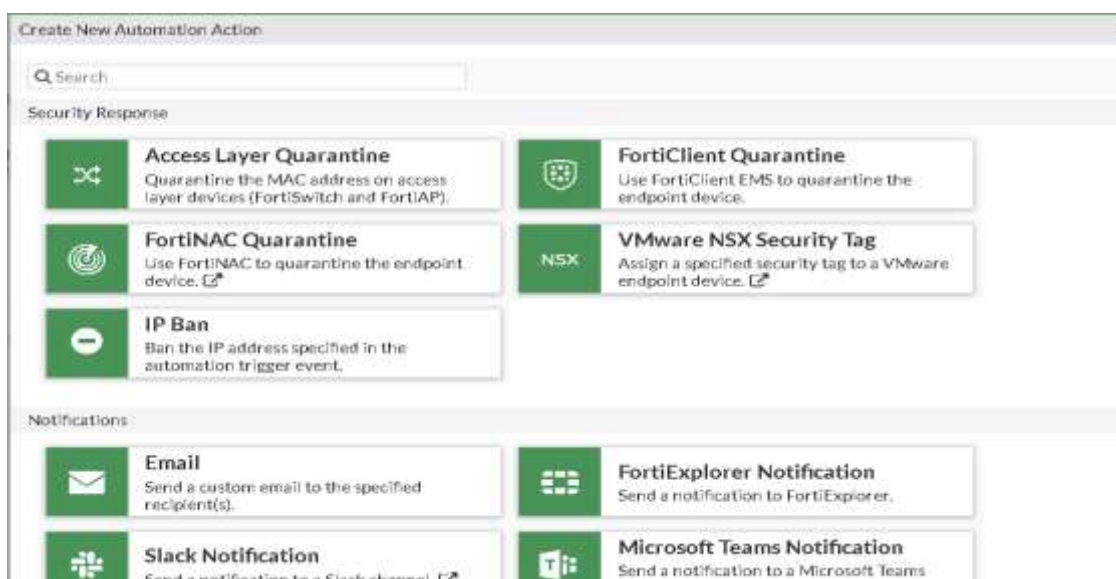


Figure 4 — Available automation actions, grouped by security response and notification

The quarantine actions differ in where isolation is enforced — switch port, endpoint agent, NAC, or hypervisor. Which one applies depends on what is deployed alongside the FortiGate, and the most effective is generally the one closest to the compromised device.

5. Assembling the Stitch

The stitch is built on a canvas: a trigger at the top, then actions beneath it, with optional delays inserted between them.

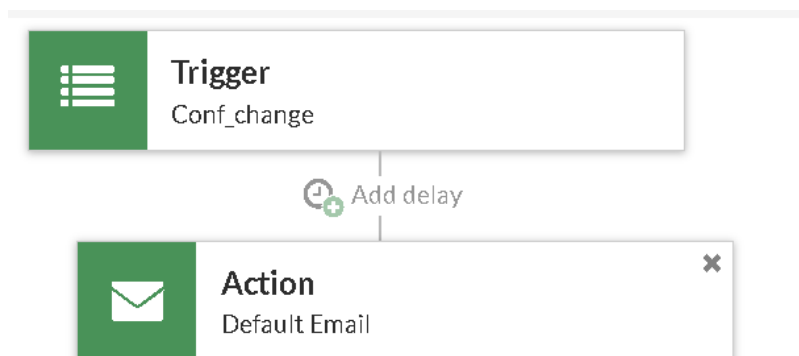


Figure 5 — The trigger and first action, with a delay insertion point between them

The delay is only meaningful in sequential mode and is specified in seconds. It exists because some responses need settling time — an action that verifies the result of the previous action will report incorrectly if it runs before that action has taken effect.

► 5.1 The CLI Script Action

A CLI script action runs arbitrary FortiOS commands when the stitch fires. It is the most flexible action available and the one that turns a stitch from a notification tool into a response tool.

The screenshot shows the 'Create New Automation Action' form for a 'CLI Script' action. The form includes fields for 'Name' (set to 'execute ping'), 'Minimum interval' (set to 0 seconds), and 'Description'. The 'CLI Script' section contains a 'Script' field with the command 'execute ping 172.16.19.44'. There are also buttons for 'Upload' and 'Record in CLI console', and a checkbox for 'Execute on Security Fabric'.

Figure 6 — A CLI script action configured to run a diagnostic ping

Field	Purpose
Name	Identifies the action. Actions are reusable across stitches, so the name should describe what the script does
Minimum interval	Rate limit, in seconds. The action will not run again until this interval has elapsed, regardless of how often the trigger fires
Script	The commands to execute. Here, a ping against a target address to confirm reachability at the moment the event occurred

Field	Purpose
Administrator profile	The privilege context the script runs under. The script can only do what this profile permits
Execute on Security Fabric	Runs the script across all FortiGates in the fabric rather than only the local device

► 5.2 Two Fields Worth Setting Deliberately

Minimum interval. Left at zero, a trigger that fires repeatedly runs the action every time. On a noisy trigger this produces a notification flood that trains the recipients to ignore it, or a script that runs hundreds of times in a minute. A non-zero interval is the difference between an alert that is read and one that is filtered.

Administrator profile. The script inherits this profile's permissions, so it should be scoped to what the script actually needs rather than defaulted to full access. A stitch is an unattended execution path into the CLI — granting it more privilege than the task requires widens the blast radius of a mistake in the script.

6. The Completed Stitch

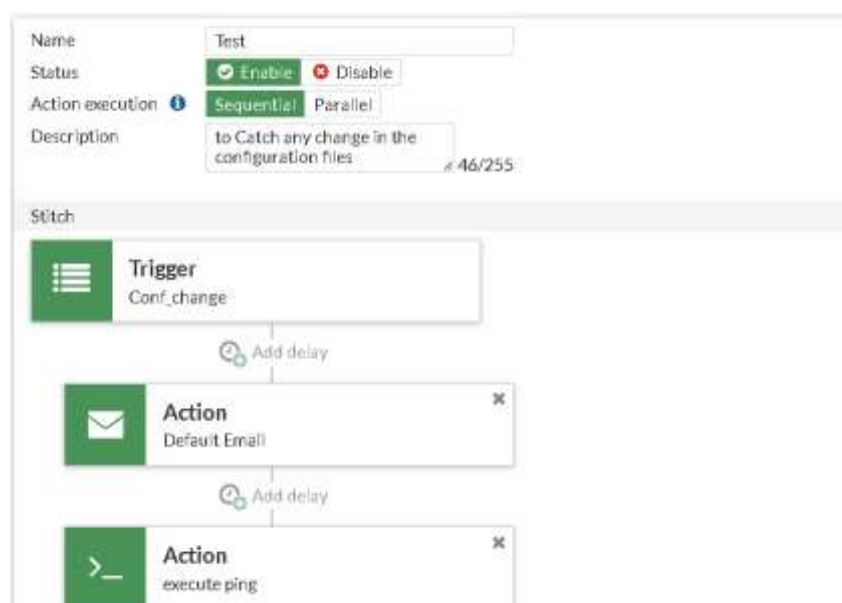


Figure 7 — The finished stitch: a configuration change triggers an email, then a diagnostic script

The stitch watches for the `Conf_change` trigger and runs two actions in sequence — an email notification followed by the ping script. Sequential execution is appropriate here because the notification should be dispatched first; if the script were to fail, the record of the event has already been sent.



Figure 8 — The stitch as it appears in the automation list, enabled and applied to all FortiGates

The final column is the hit counter, showing how many times the stitch has fired — the fastest way to confirm a stitch is live, and to spot one firing far more often than expected.

7. Verification

Testing a stitch means causing the trigger condition deliberately. The Configuration Change trigger is convenient for this because it can be raised on demand and carries no risk — unlike a compromised-host trigger, which cannot be produced safely on a production appliance.

Date/Time	Level	User	Message	Log Description
Wednesday	INFO	admin	Add system.automation-stitch Test	Object attribute configured
Wednesday	INFO	admin	Edit system.automation-action DefaultEmail	Object attribute configured
Wednesday	INFO	admin	Add system.automation-trigger Conf_change	Object attribute configured
Wednesday	INFO	admin	Add system.automation-trigger Test_Change	Object attribute configured
Wednesday	INFO		Performance statistics average CPU: 47, memory: 22, concurre...	System performance statistics

Figure 9 — Event log entries recording the automation objects as they were created

The log records each change made while building the stitch — the stitch, the trigger, and the actions — attributed to the administrator that made it. These entries are both the audit trail and the trigger source, which is why this trigger is self-demonstrating: creating the automation generates the events it watches for.

► 7.1 The Stitch Firing

Changing the appliance hostname raised the trigger and produced the full chain in the event log, read from the bottom upward.

Wednesday	ERROR	system	Failed to send alert email from notification.fortinet.net to (abdelrhman.nabil@fortinet.com)	Alert email send status failed
Wednesday	INFO	system	stitch:Test is triggered.	Automation stitch triggered!
Wednesday	INFO	admin	Configuration is changed in the admin session	Configuration changed

Figure 10 — The complete chain: configuration change, stitch triggered, email action attempted

Order	User	Event	Meaning
1	admin	Configuration is changed in the admin session	The trigger condition occurred
2	system	stitch:Test is triggered	The stitch matched the event and began executing
3	system	Failed to send alert email	The first action ran and reported its outcome

This confirms the automation is wired correctly. The trigger fired on a real configuration change, the stitch matched it without intervention, and the action chain began executing — each step attributed and timestamped.

► 7.2 The Failed Email

The third entry reports that the alert email could not be sent. This is an expected result on this appliance rather than a fault in the stitch: the VM is unlicensed, so it has no working path through Fortinet's notification service and no configured SMTP relay of its own.

Why this is a useful result. The log distinguishes clearly between two failures that are easy to confuse. **The stitch worked.** The trigger fired, the stitch matched, and the action was dispatched — that is entry 2, and it is the part being tested. What failed is the delivery mechanism behind the action, which is a mail configuration problem and would be identical for any other feature that sends email. Had the log shown no stitch entry at all, the diagnosis would be the opposite and would point at the trigger or the stitch status.

This is also the argument for testing notification stitches before relying on them. A stitch can be configured perfectly and still deliver nothing, and the only way to discover that is to fire it deliberately rather than during an incident.

8. Summary

Component	Details
Trigger	Configuration Change — raised by any admin session altering the running config
Actions	Email notification, then a CLI script running a diagnostic ping
Execution mode	Sequential — notification dispatched before the script runs
Scope	All FortiGates
Verification	Event log shows config change, stitch triggered, and action outcome in sequence
Known limitation	Email delivery failed — unlicensed VM with no SMTP path; the stitch itself executed correctly